

# Class 12

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## Class 12 Homework

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Reading the dataset

```
data <- read.table("rs8067378_ENSG00000172057.6.txt", header = TRUE, sep = "", stringsAsFactors = FALSE)
names(data)
```

```
[1] "sample" "geno"   "exp"
```

Sample Size for each genotype:

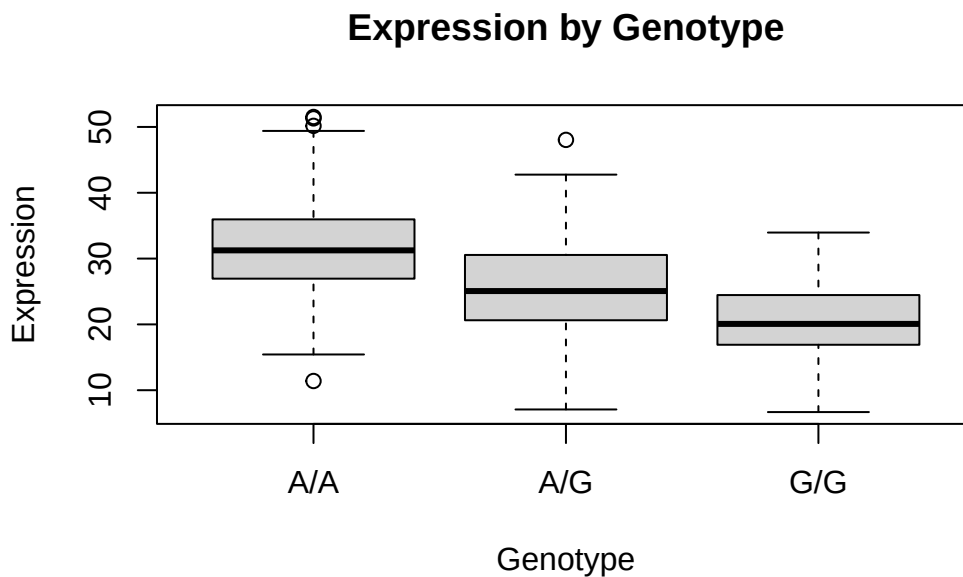
```
table(data$geno)
```

```
A/A A/G G/G  
108 233 121
```

**Q14: Generate a boxplot with a box per genotype, what could you infer from the relative expression value between A/A and G/G displayed in this plot? Does the SNP effect the expression of ORMDL3? Hint: An example boxplot is provided overleaf – yours does not need to be as polished as this one.**

Boxplot for each genotype:

```
bp <- boxplot(exp ~ geno, data = data,  
              main = "Expression by Genotype",  
              xlab = "Genotype",  
              ylab = "Expression")
```



```
bp$stats[3, ]
```

```
[1] 31.24847 25.06486 20.07363
```

Median expression of each genotype:

```
bp$stats[3, ]
```

```
[1] 31.24847 25.06486 20.07363
```

As shown by the boxplot, the A/A genotype has the highest median expression of around 30, the A/G has the middle expression of around 25, and G/G has the lowest median expression of around 20. Though there is a bit of overlap, there is a consistent decline in expression the more G alleles the genotype has.

This suggests that the SNPS does affect the expression of ORMDL\_3 with the presence of the G allele reducing expression. Individuals with G/G genotype consistently show lower expression, while those with A/A show the highest.

Summary of data:

```
summary(data)
```

|          | sample | geno    | exp            |
|----------|--------|---------|----------------|
| HG00096: | 1      | A/A:108 | Min. : 6.675   |
| HG00097: | 1      | A/G:233 | 1st Qu.:20.004 |
| HG00099: | 1      | G/G:121 | Median :25.116 |
| HG00100: | 1      |         | Mean :25.640   |
| HG00101: | 1      |         | 3rd Qu.:30.779 |
| HG00102: | 1      |         | Max. :51.518   |
| (Other): | 456    |         |                |